which are solutions to

$$\dot{z}_i(t) = \frac{1}{Z_{\beta,i}(t)} \sum_{j=1}^{n} e^{\beta\left\langle Qe^{tV} z_i(t), Ke^{tV} z_j(t)\right\rangle} V(z_j(t) - z_i(t)) \tag{9.3}$$

for $i \in [n]$ and $t \geqslant 0$, where

$$Z_{\beta,i}(t) = \sum_{k=1}^{n} e^{\beta\langle Qe^{tV} z_i(t), Ke^{tV} z_k(t)\rangle},$$

whereas the initial condition remains the same, namely $x_i(0) = z_i(0)$. It is crucial to notice that the coefficients $A_{ij}(t)$ (see (2.5)) of the self-attention matrix for the rescaled particles $z_i(t)$ are the same as those for the original particles $x_i(t)$. The weight $A_{ij}(t)$ indicates the strength of the attraction of $z_i(t)$ by $z_j(t)$. In [GLPR24] we show that the rescaled particles $z_i(t)$ cluster toward well-characterized geometric objects as $t \to +\infty$ for various choices of matrices (Q, K, V). Our results are summarized in Table 1 below, whose first two lines are discussed thereafter.

V	K and Q	Limit geometry	Result in [GLPR24]
$V = I_d$	$Q^\top K \succ 0$	vertices of convex polytope	Theorem 3.1
$\lambda_1(V) > 0$, simple	$\langle Q\varphi_1, K\varphi_1\rangle > 0$	union of 3 parallel hyperplanes	Theorem 4.1
V paranormal	$Q^\top K \succ 0$	polytope $\times$ subspaces	Theorem 5.1
$V = -I_d$	$Q^\top K = I_d$	single cluster at origin*	Theorem C.5

Table 1. Summary of the clustering results of [GLPR24]. *All results except for the case $V = -I_d$ hold for the time-scaled dynamics (9.3).

When $V = I_d$, outside from exceptional situations, all particles cluster to vertices of some convex polytope. Indeed, since the velocity $\dot{z}_i(t)$ is a convex combination of the attractions $z_j(t) - z_i(t)$, the convex hull $\mathcal{K}(t)$ of the $z_i(t)$ shrinks and thus converges to some convex polytope. The vertices of the latter attract all particles as $t \to +\infty$. When the eigenvalue with largest real part of V, denoted by $\lambda_1(V)$, is simple and positive, the rescaled particles $z_i(t)$ cluster on hyperplanes which are parallel to the direct sum of the eigenspaces of the remaining eigenvalues. Roughly speaking, the coordinates of the points $z_i(t)$ along the eigenvector of V corresponding to $\lambda_1(V)$ quickly dominate the matrix coefficients $P_{ij}(t)$ in (9.3) due to the factors $e^{tV} z_j(t)$. For more results and insights regarding clustering on $\mathbb{R}^d$, we refer the reader to [GLPR24]. We nonetheless leave the reader with the following general question:

Problem 5. *Is it possible to extend the clustering results of Table 1 to other cases of (Q, K, V)? What are the resulting limit shapes?*

9.3. **Singular dynamics.** We mention another intriguing question, whose answer would allow for a transparent geometric understanding of clustering for (9.3). Let (Q, K, V) be given $d \times d$ matrices. For $\beta > 0$, we consider the system of coupled ODEs

$$\dot{z}_i(t) = \frac{1}{Z_{\beta,i}(t)} \sum_{j=1}^{n} e^{\beta\langle Qz_i(t), Kz_j(t)\rangle} V(z_j(t) - z_i(t)), \tag{9.4}$$

where once again

$$Z_{\beta,i}(t) = \sum_{k=1}^{n} e^{\beta\langle Qz_i(t), Kz_k(t)\rangle}.$$

For any $T > 0$, and any fixed initial condition $(z_i(0))_{i\in[n]} \in (\mathbb{R}^d)^n$, as $\beta \to +\infty$, we expect that the solution to (9.4) converges uniformly on $[0, T]$ to a solution of

$$\dot{z}_i(t) = \frac{1}{|C_i(t)|} \sum_{j\in C_i(t)} V(z_j(t) - z_i(t)) \tag{9.5}$$

where

$$C_i(t) = \Big\{ j \in [n] \colon \langle Qz_i(t), Kz_j(t)\rangle \geqslant \langle Qz_i(t), Kz_k(t)\rangle \quad \text{for all } k \in [n] \Big\}. \tag{9.6}$$

However, defining a notion of solution to (9.5)–(9.6) is not straightforward, as illustrated by the following example.

Example 9.3. *Suppose $d = 2$, $n = 3$. Let $Q = K = V = I_d$ and $z_1(0) = (1, 1)$, $z_2(0) = (-1, 1)$, $z_3(0) = (0, 0)$. Consider the evolution of these particles through (9.5)–(9.6). The points $z_1(t)$ and $z_2(t)$ do not move, because it is easily seen that $C_i(t) = \{i\}$ for $i \in \{1, 2\}$. On the other hand, the point $z_3(t)$ can be chosen to solve either of three equations: $\dot{z}_3(t) = z_1(t) - z_3(t)$, or $\dot{z}_3(t) = z_2(t) - z_3(t)$, or even $\dot{z}_3(t) = \frac{1}{2}(z_1(t) + z_2(t)) - z_3(t)$. In any of these cases, both (9.5) and (9.6) remain satisfied for almost every $t \geqslant 0$.*

It is possible to prove the existence of solutions to (9.5)–(9.6) defined in the sense of Filippov[15]: for this, we can either use a time-discretization of (9.5)–(9.6), or use a convergence argument for solutions to (9.4) as $\beta \to +\infty$. Uniqueness however does not hold, as illustrated by Example 9.3. This naturally leads us to the following question:

Problem 6. *Is it possible to establish a selection principle (similar to viscosity or entropy solutions) for solutions to (9.5)–(9.6) which allows to restore uniqueness? In the affirmative, is it possible to revisit the clustering results of [GLPR24] and Problem 5 in the setting of (9.5)–(9.6)?*

9.4. **Diffusive regularization.** We believe that (9.5)–(9.6) is also an original model for collective behavior. There are some similarities in spirit with methods arising in *consensus based optimization* (CBO for short), [PTTM17, CJLZ21]. With CBO methods, one wishes to minimize a smooth and bounded, but otherwise arbitrary function $f : \mathbb{R}^d \to \mathbb{R}$ by making use of the Laplace method

$$\lim_{\beta\to+\infty} \left(-\frac{1}{\beta} \log \int_{\mathbb{R}^d} e^{-\beta f(x)} \, \mathrm{d}\rho(x) \right) = \inf_{x\in\mathrm{supp}(\rho)} f(x),$$

which holds for any fixed $\rho \in \mathcal{P}_{\mathrm{ac}}(\mathbb{R}^d)$. This is accomplished by considering a McKean-Vlasov particle system of the form

$$\mathrm{d}x_i(t) = -\lambda(x_i(t) - v_f)H^\epsilon(f(x_i(t)) - f(v[\mu_n(t)]))\,\mathrm{d}t + \sqrt{2}\sigma|x_i(t) - v[\mu_n(t)]|\,\mathrm{d}W_i(t)$$

for fixed $\beta > 0$, with drift parameter $\lambda > 0$ and noise parameter $\sigma \geqslant 0$; H^ϵ is a particular smoothed Heaviside function, and $\mu_n(t)$ is the empirical measure of the

[15]We thank Enrique Zuazua for this suggestion.

particles. The point $v[\mu] \in \mathbb{R}^d$ is a weighted average of the particles:

$$v[\mu] = \frac{1}{Z_{\beta,\mu}} \int_{\mathbb{R}^d} e^{-\beta f(x)} x \, \mathrm{d}\mu(x)$$

where $Z_{\beta,\mu} = \int_{\mathbb{R}^d} e^{-\beta f(x)} \, \mathrm{d}\mu(x)$. Morally speaking, particles which are near a minimum of f have a larger weight. The drift term is a gradient relaxation (for a quadratic potential) towards the current weighted average position of the batch of particles. The diffusion term is an exploration term whose strength is proportional to the distance of the particle from the current weighted average. Results of convergence to a global minimizer do exist, under various smallness assumptions on the initial distribution of the particles, and assumptions on the relative size of the coefficients. They rely on the analysis of the associated Fokker-Planck equation, see [CJLZ21, CD22], and also [FHPS21] for the analog on $\mathbb{S}^{d-1}$. We point out that similarities are mainly in spirit—these results and analysis are inapplicable to our setting because there is no analog for $f(x)$—. Nonetheless, they do raise the following interesting question:

Problem 7. *What can be said about the long-time limit of Transformers with a noise/diffusion term of strength $\sigma > 0$?*

The question is of interest for any of the Transformers models presented in what precedes.

10. Approximation, control, training

Understanding the *expressivity*, namely the ability of a neural network to reproduce any map in a given class (by tuning its parameters), is essential. Two closely related notions reflect the expressivity of neural networks: *interpolation*—the property of exactly matching arbitrarily many input and target samples—and *(universal) approximation*—the property of approximating input-target functional relationships in an appropriate topology—. We refer the reader to [CLLS23] for a primer on the relationship between these two notions in the contex of deep neural networks.

For discrete-time Transformers, universal approximation has been shown to hold in [YBR+19], making use of a variant of the architecture with translation parameters and letting the number of layers go to infinity; see also [ADTK23, JL23] and the review [JLLW23].

In the context of flow maps (from $\mathbb{R}^d$ to $\mathbb{R}^d$), it is now well understood that interpolation and approximation reflect the *controllability* properties of the system. The transfer of control theoretical techniques to the understanding of expressivity has borne fruit, both in terms of controllability results [AS22, CLT20, TG22, LLS22, RBZ23, VR23, CLLS23] and optimal control insights [LCT18, EGPZ20, GZ22, EYG23]. We refer the reader to [AL24, AG24, FdHP24] for the first universal approximation results for Transformers, viewed as measure-to-measure maps, using control theoretic tools.

Besides approximation, understanding the training dynamics of Transformers is another major challenge which we haven't covered herein. As it is impossible to reference all works on this burgeoning topic, we refer the interested reader to [TLTO23, ACDS23, DGTT24] and references therein.